Model metrics (estimate [95% bootstrap CI])

metric	RF_Luke	XGB_Thomas
brier	0.1021 [0.0963, 0.1078]	0.1014 [0.0955, 0.1072]
log_loss	0.3158 [0.2992, 0.3323]	0.3109 [0.2943, 0.3274]
brier_skill	0.5609 [0.5279, 0.5945]	0.5637 [0.5301, 0.5970]
roc_auc	0.9287 [0.9189, 0.9377]	0.9304 [0.9210, 0.9391]
average_precision	0.8991 [0.8749, 0.9192]	0.9000 [0.8767, 0.9203]
max_youden_ks	0.6907 [0.6705, 0.7159]	0.6935 [0.6706, 0.7166]
cal_intercept	-0.0601 [-0.1330, 0.0142]	0.0107 [-0.0690, 0.0909]
cal_slope	1.0526 [0.9942, 1.1139]	0.9781 [0.9289, 1.0296]
ece	0.0101 [0.0092, 0.0205]	0.0109 [0.0082, 0.0203]
accuracy	0.8511 [0.8407, 0.8611]	0.8511 [0.8410, 0.8611]
balanced_accuracy	0.8322 [0.8171, 0.8471]	0.8316 [0.8162, 0.8467]
tpr_recall	0.7611 [0.7320, 0.7899]	0.7578 [0.7290, 0.7870]
tnr_specificity	0.9034 [0.8925, 0.9133]	0.9054 [0.8945, 0.9156]
ppv_precision	0.8207 [0.7929, 0.8456]	0.8232 [0.7964, 0.8485]
npv	0.8668 [0.8569, 0.8760]	0.8654 [0.8559, 0.8750]
f1	0.7898 [0.7638, 0.8147]	0.7891 [0.7629, 0.8140]
mcc	0.6759 [0.6469, 0.7037]	0.6758 [0.6465, 0.7043]
cohen_kappa	0.6747 [0.6458, 0.7026]	0.6744 [0.6452, 0.7029]
youden_j	0.6645 [0.6342, 0.6943]	0.6632 [0.6323, 0.6935]
expected_cost	0.1489 [0.1389, 0.1593]	0.1489 [0.1389, 0.1590]
cost_vs_trivial	0.4052 [0.3582, 0.4547]	0.4050 [0.3582, 0.4532]
brier_MCB	0.0015	0.0014
brier_DSC	0.1319	0.1324
brier_UNC	0.2325	0.2325
logloss_MCB	0.0053	0.0050
logloss_DSC	0.3471	0.3517
spiegelhalter_z	-3.4644	2.3857
spiegelhalter_p	<0.001	0.017
n_extreme_probs	0	0
n_confident_misses	0	0
roc_auc_delong_se	0.0019	0.0018

Paired comparisons (difference = first model minus second)

RF_Luke minus XGB_Thomas

metric	difference [95% CI]	test	p	p (Holm)
brier	0.00067 [-0.00053, 0.00191]	Diebold-Mariano	0.090	0.090
log_loss	0.00490 [0.00153, 0.00836]	Diebold-Mariano	<0.001	<0.001
roc_auc	-0.00169 [-0.00359, 0.00008]	DeLong	0.002	0.002
average_precision	-0.00090 [-0.00347, 0.00161]	paired bootstrap	0.458	0.458
accuracy	-0.00006 [-0.00424, 0.00408]	McNemar mid-p	0.969	0.969
balanced_accuracy	0.00066 [-0.00406, 0.00497]	paired bootstrap	0.822	0.822
f1	0.00067 [-0.00542, 0.00637]	paired bootstrap	0.874	0.874
mcc	0.00012 [-0.00904, 0.00919]	paired bootstrap	0.972	0.972
expected_cost	0.00006 [-0.00408, 0.00424]	paired bootstrap	0.942	0.942
cal_slope	0.07451 [0.05007, 0.09936]	paired bootstrap	<0.001	<0.001
ece	-0.00079 [-0.00735, 0.00909]	paired bootstrap	0.834	0.834

Per-group metrics (by hotel)

Resort Hotel (n = 5,785, prevalence 0.269)

metric	RF_Luke	XGB_Thomas
brier	0.0969 [0.0915, 0.1030]	0.0949 [0.0893, 0.1010]
log_loss	0.3071 [0.2913, 0.3275]	0.2983 [0.2818, 0.3179]
roc_auc	0.9233 [0.9122, 0.9340]	0.9262 [0.9156, 0.9364]
cal_intercept	-0.0333 [-0.1259, 0.0550]	0.0436 [-0.0502, 0.1385]
cal_slope	1.1355 [1.0421, 1.2345]	1.0507 [0.9772, 1.1321]
accuracy	0.8598 [0.8484, 0.8706]	0.8598 [0.8480, 0.8696]
mcc	0.6352 [0.6002, 0.6705]	0.6339 [0.5994, 0.6665]
expected_cost	0.1402 [0.1294, 0.1516]	0.1402 [0.1304, 0.1520]

City Hotel (n = 12,118, prevalence 0.415)

metric	RF_Luke	XGB_Thomas
brier	0.1046 [0.0970, 0.1124]	0.1045 [0.0969, 0.1121]
log_loss	0.3199 [0.2982, 0.3420]	0.3169 [0.2955, 0.3380]
roc_auc	0.9284 [0.9157, 0.9392]	0.9295 [0.9177, 0.9402]
cal_intercept	-0.0730 [-0.1695, 0.0270]	-0.0047 [-0.0977, 0.0991]
cal_slope	1.0106 [0.9343, 1.0874]	0.9413 [0.8811, 1.0024]
accuracy	0.8469 [0.8347, 0.8603]	0.8470 [0.8342, 0.8598]
mcc	0.6826 [0.6484, 0.7158]	0.6828 [0.6440, 0.7149]
expected_cost	0.1531 [0.1397, 0.1653]	0.1530 [0.1402, 0.1658]